

Creating and Sharing Genomic Annotation Modules with AI: The Just-DNA Ecosystem

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Abstract

Genome annotation depends on curated knowledge that changes as papers and databases publish new variant-trait associations. The Just-DNA ecosystem stores this knowledge in genomic annotation modules: versioned data packages that describe which genotype a claim concerns, what evidence supports it, and how a consumer should present it. A module can be written and edited manually. An AI assistant can also draft one from a paper, a variant list, or a description of a topic, but its output remains an editable data artifact rather than generated analysis code.

We present the Just-DNA ecosystem and the path a module follows through it. `just-dna-format` defines the schema; `just-dna-enricher` records source-derived facts; `just-dna-compiler` validates and compiles the files. `just-module-creator`, an open-source plugin for Claude Code and Codex, is the authoring entry point: it exposes these packages as typed tools so that both AI-assisted and manually edited modules use the same live schema. An author can keep the result in a local module store, rehearse publication in a deletable test registry, or publish a version to the production registry. Just-DNA-Lite later filters a local VCF and joins selected modules to matching genotypes; its internal design is the subject of a companion platform paper [Anonymous, 2026].

The language model helps with reading and drafting, while deterministic software handles schema validation, compilation, genome processing, and scoring. This paper describes the module lifecycle from source material to shared store and local use. It proposes a protocol for evaluating variant recovery, citation identity, and effect-direction agreement in AI-assisted creation, and reports catalog statistics from the first eight published modules. The software is open-source and intended for research use only.

1 Introduction

Suppose a researcher wants a recent lactose-tolerance study to inform a genomic report. The researcher can transcribe the relevant variants and evidence into tables, or ask an AI assistant to prepare a first draft. In either case, the result is the same kind of object: a module that can be edited, checked, versioned, and shared. The researcher may keep it in a local store while it is being reviewed, publish it to a test registry for rehearsal, and later release an accepted version through the production registry. When a user selects that module in Just-DNA-Lite, the application joins its records to matching genotypes in a local VCF and includes the resulting annotations in a report [Anonymous, 2026].

This separation between knowledge and sample processing is central to the design. A paper may report an association with lactose tolerance, drug response, or another trait, but a computational pipeline needs an unambiguous variant, the genotype to which the claim applies, the genome build and allele orientation, a structured conclusion, and a trace back to the evidence. A *module* is a versioned collection of those records. It contains no sample data and no executable analysis code. The same compiled module can be applied to many genomes.

The sample follows a different path. A VCF often contains millions of variant calls. Just-DNA-Lite normalizes their representation, applies configured quality filters, and joins the retained calls to selected modules and reference data. That pipeline and its benchmarks are described in the companion platform paper [Anonymous, 2026]; polygenic risk scores follow a parallel numerical path through `just-prs`.

A general-purpose assistant is useful during curation, but it does not provide this structure by itself. Asked to “make me a lactose-tolerance module,” it may return rsIDs, an explanation, and

a new script that searches a VCF. The answer can contain a fabricated identifier, a reversed allele, or a real PMID that names the wrong paper [Ji et al., 2023, Gao et al., 2023]. Language models can also generate plausible but non-existent gene names or variant identifiers [Jin et al., 2024b]. Generated code may bypass the established filters and validators or depend on an unstated environment. The next run can produce a different implementation because the implementation was generated as part of the answer.

The Just-DNA ecosystem gives that assistant a narrower and more useful job. `just-module-creator` helps it read sources, prepare module files, run checks, and present unresolved decisions to the author. The files use schemas from `just-dna-format`; `just-dna-enricher` records facts from external sources; `just-dna-compiler` validates and compiles them; and `just-dna-registry` stores published versions. Manual authors use the same files and tools. Just-DNA-Lite remains responsible for sample-level computation [Anonymous, 2026].

This paper makes four contributions:

1. It describes the module contract shared by manual editors, AI-assisted authoring tools, versioned stores, and the genome-analysis consumer.
2. It presents `just-module-creator`, an authoring plugin that exposes the public capabilities of `just-dna-format`, `just-dna-compiler`, `just-dna-enricher`, and `just-dna-registry` as typed MCP tools [Anthropic, 2024]. The plugin reads the installed schema instead of carrying a copy.
3. It defines how modules move between a local store, a deletable test registry, and the production registry while preserving provenance and decisions that still require review.
4. It proposes an evaluation protocol for variant recovery, citation identity, and effect-direction agreement, and reports catalog statistics from the first eight published modules across five independent namespaces.

2 Related work

Genome annotation and catalogs. A variant call records what differs between a sample and a reference genome. Annotation adds information about that call, such as its predicted consequence, known clinical classification, reported trait association, or the conclusion assigned to a particular genotype. VEP, ANNOVAR, and OpenCRAVAT/OakVar perform this kind of lookup by joining a callset to established resources [McLaren et al., 2016, Wang et al., 2010, Pagel et al., 2020]. ClinVar and the GWAS Catalog publish records that can supply annotation content [Landrum et al., 2018, Sollis et al., 2023]. HGMD derives variant–disease associations from the literature through manual review, though the public version updates infrequently and does not include explicit pathogenicity calls [Stenson et al., 2020]. ClinGen provides expert variant curation within the ACMG/AMP framework [Rehm et al., 2015, Richards et al., 2015]. PharmGKB and CPIC supply pharmacogenomic annotations including diplotype-level clinical recommendations [Whirl-Carrillo et al., 2012]. The GA4GH Genomic Knowledge Standards define interoperable representations for variant annotations [Global Alliance for Genomics and Health, 2022]. The problem addressed here begins before the join: selected evidence must be turned into a reviewable, machine-checkable collection that can be distributed and applied consistently.

Table 1 compares the authoring capabilities of these tools with `just-module-creator`.

Biomedical text mining and variant extraction. Conventional named entity recognition systems such as tmVar3 and PubTator3 have improved gene and variant tagging in biomedical text [Wei et al., 2022, 2024]. LitVar 2 links literature paragraphs with variant records

Table 1: Qualitative comparison of authoring and sharing capabilities. The comparison focuses on the *authoring* side of the boundary: how new annotation content is created, validated, and distributed. It does not compare runtime annotation speed, which is the subject of the companion platform paper [Anonymous, 2026].

	<i>Versioned modules</i>	<i>Schema validation</i>	<i>Provenance tracking</i>	<i>AI-assisted authoring</i>	<i>Community registry</i>
VEP / ANNOVAR	–	–	–	–	–
OpenCRAVAT / OakVar	✓	✓	–	–	✓
ClinVar / ClinGen	✓	✓	✓	–	✓
HGMD	✓	–	✓	–	–
PharmGKB / CPIC	✓	✓	✓	–	–
GA4GH GKS	✓	✓	✓	–	–
PubMind-DB	–	✓	✓	✓	✓
Raw LLM	–	–	–	✓	–
just-module-creator	✓	✓	✓	✓	✓

from external databases, accelerating knowledge retrieval [Allot et al., 2023]. However, these tools are largely limited to entity-level extraction and do not recover relational information such as variant-specific pathogenicity across disease contexts or the experimental evidence supporting a classification. PubMind applies instruction-tuned LLMs to extract variant–disease–pathogenicity associations directly from over 41 million PubMed abstracts and 5.4 million full-text articles, producing a database of approximately 1.3 million unique variants with contextual annotations [Wang and Wang, 2026]. These systems demonstrate that LLM-based literature mining can recover structured variant knowledge at scale, but their output is a flat database rather than a versioned, schema-validated module that an annotation consumer can install and apply to a genome.

Factuality and tool use. Language models can produce fluent text around a false identifier or citation [Ji et al., 2023]. Benchmarks on genomic question-answering show that models hallucinate gene names, variant identifiers, and functional annotations [Jin et al., 2024b]. Retrieval-augmented approaches and tool use reduce the need to improvise when a model can call a typed operation instead [Schick et al., 2023, Jin et al., 2024a]. The Model Context Protocol provides a standard for connecting models to external tools [Anthropic, 2024]. Biomedical MCP services can search literature, ontologies, and variant databases. just-module-creator uses the same general approach, then carries the result into module files that the rest of the Just-DNA pipeline can inspect and reuse.

Agent orchestration. Many scientific-agent systems assign research and review to several model instances [Hong et al., 2023]. The plugin described here does not require its own orchestration runtime. Claude Code or Codex supplies the agent, while the plugin supplies the tools and the written workflow. This paper evaluates that authoring path, not a particular choice of language model or agent-team topology.

3 Method

3.1 How modules move through the ecosystem

Module authoring and sample analysis remain separate until Just-DNA-Lite joins a compiled module to a local genome. On the authoring path, a person may edit the files directly or use an AI assistant to draft them. Both routes then pass through the same enrichment and compilation steps before the module enters a local or shared store. Figure 1 shows the two paths and the boundary between them.

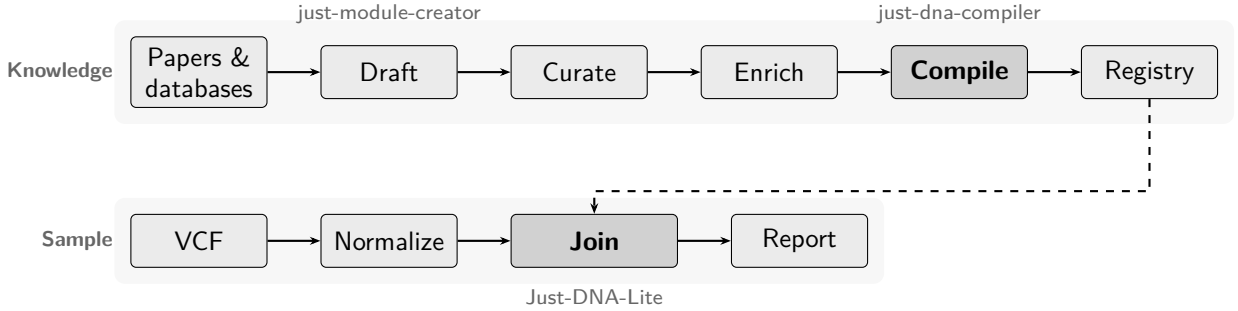


Figure 1: The two paths of the Just-DNA ecosystem. The knowledge path (top) produces a compiled module from sources; the sample path (bottom) joins that module to a local genome. The compiler is deterministic and offline. The dashed arrow marks the point where the two paths meet: a published or locally installed module is consumed by Just-DNA-Lite.

Table 2: The module lifecycle and the two sample-processing paths.

Path	Stages
Module lifecycle	papers and databases → manual editing or AI draft → enrichment and compilation → local store or registry
Sample annotation	VCF → normalization and quality filtering → joins with selected modules and reference data → annotated tables and report

On the sample path, normalization gives equivalent variants a consistent representation and computes the genotype used by later joins. Configured quality filters can remove calls that fail VCF filter status, depth, or quality criteria. The pipeline writes the normalized data to Parquet, so each selected module can use a streaming join instead of independently parsing the original VCF. The companion platform paper [Anonymous, 2026] describes the sample path in detail, including annotation speed benchmarks (a whole-genome VCF annotated in under 40 seconds) and polygenic risk score computation via **just-prs**.

Within **just-module-creator**, the language model participates only in the knowledge path. It can search and read sources, draft module rows, and help review disagreements. VCF filtering, module joins, and polygenic scoring remain ordinary software steps with explicit inputs and repeatable code.

3.2 The module artifact

A just-dna annotation module begins as a directory. It always contains `module_spec.yaml` and at least one recognized table. The table records the thing being annotated, such as a genotype, a diplotype, or a measured quantity. Evidence and machine-produced sidecars are stored separately. The compiler turns this directory into parquet files and a content-addressed `manifest.json` [DNA Seq contributors, 2026].

The authored files are ordinary YAML, CSV, Markdown, and optional image files. They can be created and revised without an AI assistant. `just-module-creator` operates on this same directory rather than introducing an AI-specific module format. The module describes how a genotype or measurement should be interpreted if a consumer encounters it. Just-DNA-Lite, or another compatible consumer, supplies the genome or measurement later.

The table kind follows the subject of the claim. For example, in format 0.6.6 used for this manuscript, a single-variant claim belongs in `variants.csv`; evidence for those rows belongs in `studies.csv`. Pharmacogenomic diplotypes, copy-number ranges, repeat counts, and published polygenic-score identifiers use other tables. The manuscript does not reproduce their column lists. The plugin calls `describe_table` and `table_requirements`, which read the installed pydantic models. An author therefore sees the schema accepted by the compiler running in that session.

3.3 Components and responsibilities

No single package implements the whole path. Each component owns a boundary that another component can call without reimplementing its rules. Within the module toolchain, dependencies point inward: `enricher` $\rightarrow$ `compiler` $\rightarrow$ `format`. The format and compiler do not fetch remote data. The enricher performs source lookups and writes the derived sidecars used later by the compiler. The registry publishes the compiled result, and Just-DNA-Lite consumes it in a separate application.

Table 3: Responsibilities across the Just-DNA ecosystem.

Component	Responsibility	Does not
<code>just-module-creator</code>	research and authoring workflow	process a genome
<code>just-dna-format</code>	schema and identity rules	access the network
<code>just-dna-compiler</code>	validate, compile, reverse, and sign	access the network
<code>just-dna-enricher</code>	resolve, draft, and cross-check	decide scientific meaning
<code>just-dna-registry</code>	publish, discover, and download modules	author module rows
Just-DNA-Lite	filter VCFs, join annotations, and produce reports	define module schema or compiler rules

Of these components, `just-module-creator` is the agent-facing authoring entry point examined most closely in this paper. It is distributed as an application whose public contract is the MCP tool surface and the accompanying workflow documents. The same source tree ships plugin manifests for Claude Code and Codex. Both hosts launch the same server and load the same skills.

3.4 Compiler behaviour belongs to the compiler

Within one compiler version, `just-dna-compiler` tests that a valid specification can be compiled, reversed, and compiled again without changing its authored content signature. `just-module-creator` calls that compiler and pins the resolution setting.¹ It does not implement a second compiler or present the upstream round-trip tests as evidence that an assistant extracted a paper correctly.

This separation avoids a circular evaluation. If an assistant writes a wrong claim in valid syntax, the compiler may preserve it perfectly. Reproducible output shows that the compiler

¹The plugin always calls the compiler with `resolve_with_ensembl=True`. The parameter name is misleading: setting it to false disables all resolution, including an injected `resolution.csv`, and the compiler can then succeed with null coordinates that cannot match a VCF.

kept the input stable; it does not show that the input agrees with the literature.

3.5 Tools available to the assistant

The MCP tools are grouped by task: authoring, research, checks, enrichment, comparison, and registry work. The default configuration lists the full surface. An optional layered configuration initially lists the core authoring loop and a `toolbox` command that reveals the other groups. Tool search can reduce the listing further. These discovery options change what the agent sees in its context; they do not create a separate implementation of the workflow.

Registry writes require a token, with one necessary exception: `registry_register` creates the token and therefore cannot require one in advance. Tools whose cost depends on a large source corpus state that cost in their descriptions.

Literature and enrichment requests share a pacing gate, including the NCBI budget used by several lookups. The literature search fills a gap left by the enricher. The enricher can verify a PMID that an author already has, but it does not search for the paper. `literature_search` returns titles so the assistant can check identity. The existence of a PMID alone says nothing about whether it is the intended article.

3.6 How a person enters the workflow

A person usually knows what they want to accomplish, not which internal stage performs it. They may want to create a module, understand an inherited directory, find a paper, revise a published module, or decode an error message. The command menu uses those intentions as its entries.

Command	Purpose
<code>/create-module</code>	begin or continue creating a module
<code>/module-status</code>	inspect a directory and identify the next decision
<code>/module-revise</code>	begin another pass over an existing module
<code>/find-evidence</code>	find and read evidence for a row
<code>/module-publish</code>	rehearse and then publish
<code>/module-symptom</code>	explain a message from the toolchain
<code>/module-install-local</code>	install a compiled module without a catalog

The first version of the menu exposed twenty stage names. That forced a person to choose between terms such as `module-curate` and `module-enrich` before the system had explained either one. The seven commands now route to thirteen guides. A guide contains the procedure for one stage and is loaded by path when the assistant reaches that stage.

`/create-module` is therefore a router. It examines what the author has already provided and selects the next stage. If the request shows that the person first needs an explanation, the router loads the overview guide. The overview is not a menu command because a new author would not know its internal name.

3.7 The authoring journey

The workflow follows the module from an idea to one of its destinations:

origin → scaffold → draft → curate → enrich → check → compile → close →
rehearse → publish

The origin determines where the work begins. A source such as ClinVar, CPIC, or ClinPGx may already publish rows, in which case a drafting tool can prepare a partial table. A paper or a free-text request instead begins with evidence search and manual authoring. Scaffolding creates the files and placeholders required for the selected table kind.

Curation follows drafting because a drafted row may still contain decisions that the source cannot make for the module. The author or assistant must decide which genotype the claim concerns, how a weight should be interpreted, and how the conclusion should be worded. The workflow presents these decisions rather than silently choosing values that merely satisfy the schema.

Enrichment resolves identifiers and writes sidecars such as `resolution.csv` and `literature.csv`. These files record the source answers used for later offline compilation. Compilation itself does not use the network.

After compilation, `close` records a statement in `verification.json` and binds it to the authored files. Editing those files invalidates the closure. In the current 0.x format a module can still compile and publish without a closure, but it carries a warning. Closing is an attributed declaration of completion. It does not prove that the module is correct.

Most modules return to this workflow. A later source release, a new paper, or a reviewer's correction begins a revision pass. The router inspects the existing directory so that the assistant does not treat a revision as a blank first draft and overwrite decisions whose history matters.

3.8 Edits, evidence, and responsibility

The compiler reports problems but does not edit authored values. `just-module-creator` is the authoring application, so its workflow permits edits. That permission comes with limits.

First, tool-mediated overrides append a record to `logs/authoring.log` and preserve a reason in `provenance.json`. A general hand edit is not captured automatically. The server therefore instructs an assistant to call `record_override` after such a change. This is a known gap between the desired provenance record and what the current tools can enforce.

Second, the workflow does not fill a value from the source that will later be used to check that same value. Such a check would compare the source with itself. The same problem appears when a row uses an article title as its `provenance_quote`: the title is guaranteed to occur in the article, so the resulting match provides no evidence that anyone located support for the row's claim.

Third, disagreement with an archive requires review of both sides. An archive may lag a retraction, a meta-analysis, or a later reclassification. Replacing the module's value with the archive's value can therefore make a module worse. `record_override` stores which field differs, what the source said, who made the decision, and why the authored value should remain. The record does not turn the disagreement into a passed check.

An assistant may read retrieved full text and locate a `provenance_quote`. It must copy the passage verbatim and identify who located it in `StudyRow.curator`. This attribution helps a reviewer decide where to look closely. It does not transfer responsibility away from the human author.

3.9 Local and shared module stores

In this paper, a module store is a place from which a compiled module can be installed or discovered. A local installation is useful during authoring. The registry provides two shared stores for publication rehearsal and release.

The registry has two independent instances. Production is the catalog used by consumers. The polygon, selected with `target=test`, is a rehearsal environment where an author can delete a test publication. Writes default to the polygon. Catalog reads require an explicit target so that an author does not publish to one instance, read from the other, and mistake the missing result for a failed publication.

The production catalog is immutable. Publishing a version also claims its authored content by hash, and yanking the version does not release that claim. The workflow therefore rehearses on the polygon and asks separately before a production publish. The registry accepts the

specification, runs its own enrichment and strict compilation, and stores the artifact it produced. It does not rely on a locally claimed digest, and it does not require the author to sign the local artifact.

Publication is not the only way to use a module. `/module-install-local` installs a compiled artifact into a Just-DNA-Lite checkout so it can be tested with a local VCF. This route tests the annotation connection between the module and the consumer. A polygon publication tests the separate connection between the module and the registry.

4 Results

4.1 Production catalog

At the time of writing, the production registry holds eight published modules across 19 versions and five independent namespaces (three distinct owners). Table 4 summarizes the published modules.

Table 4: Modules published to the production registry as of August 2026. All modules target GRCh38 and were compiled with strict resolution.

Module	Variants	Studies	Genes	Versions
<code>big_five_personality</code>	330	390	185	4
<code>risk_impulsivity</code>	474	–	325	1
<code>cognitive_intelligence</code>	32	–	31	1
<code>aggression_anger</code>	28	–	22	1
<code>bodybuilding</code>	13	–	13	1
<code>placebo_response</code> (v2)	3	–	3	2
<code>placebo_response_claude</code>	3	–	3	1
<code>lactose_tolerance</code>	2	8	1	2
Total	885			19 versions

The modules span three independent namespaces (`antonkulaga`, `eric-mods`, `ksuha-dna`) from three owners. The largest module (`risk_impulsivity_snps`, 474 variants across 325 genes) and the smallest (`lactose_tolerance`, 2 variants in 1 gene) both compiled with strict resolution and fully resolved coordinates. The `big_five_personality_snps` module went through four published versions (1.0.0 through 2.1.0), illustrating the revision workflow: 1.0.0 was the initial GWAS Catalog extraction, 1.0.1 back-populated schema axes, 2.0.0 added polygenic score references, and 2.1.0 corrected the weight normalization.

Modules authored with `just-module-creator` carry `curator: ai-module-creator` in their manifest authorship records. Three modules were authored by people who are not the plugin’s developers (`eric-mods`, `ksuha-dna`), indicating that the tool surface is usable beyond the original team.

4.2 Command menu reduction

The command redesign reduced the text placed in every session from 14,688 characters for twenty entries to 3,464 characters for seven entries, a 76% reduction. A test walks the links from those seven commands and confirms that every guide remains reachable. The smaller menu therefore removes repeated prompt content without making a stage inaccessible.

4.3 The title-as-quote observation

A measurement across 33 `studies.csv` files (44,342 rows) in the upstream repository found that 3,668 rows carried a `provenance_quote` that was the article’s title verbatim. Since a title

always appears in its own full text, the quote-verification check matched on all 3,668 rows while evidencing nothing about whether anyone located support for the row’s claim. This observation led to the attribution requirement described in Section 3.8: a quote must name who located it in `StudyRow.curator` so that a reviewer can distinguish a machine-located passage from a human-read one.

4.4 Proposed evaluation protocol

The creation evaluation begins with an expert-checked fixture. Each fixture contains a free-text prompt and the `module_spec.yaml`, `variants.csv`, and `studies.csv` that the assistant should recover. The generated module is compared with the fixture on three questions:

- Variant recall is the fraction of ground-truth (*rsID*, *genotype*) rows present in the generated module.
- Citation identity requires every cited PMID to exist and to name the paper indicated by the title returned during search. Existence alone is insufficient.
- Weight-sign accuracy measures whether the generated effect direction agrees with the fixture. Magnitude is reported separately because a module weight is an authored choice and is not copied from a GWAS beta.

The development fixtures `fto_bmi` (one locus at rs1421085) and `longevity_2026` are available in the repository. A larger adjudicated set and repeated runs are needed to support a performance claim. We therefore report no recall or precision estimate in this paper and present the protocol as a framework for future evaluation.

4.5 How to read a green result

Several outputs look reassuring while answering a narrower question than a reader may expect. Strict compilation checks reproducibility and applies the compiler’s blocking rules; it does not establish biological correctness. A digest match likewise shows that bytes or authored content agree under a specified comparison.

A source timeout produces an unknown result, not a negative one and not a pass. The tools preserve this distinction with null and unknown values. The title-as-quote observation above is another illustration: a green check can mean that the instrument could not have failed rather than that it found something.

The proposed evaluation protocol counts none of these as evidence of accurate extraction. Compiler round-trip behaviour is tested by `just-dna-compiler` and is not counted again as module creation accuracy.

5 Discussion

The ecosystem’s architecture changes what an AI assistant is asked to produce. It writes module data for an existing pipeline instead of inventing the pipeline again for each question. The same compiler and schema are used for every module, and the sample is normalized once before selected annotations are joined through the established columnar implementation. The language model may still misunderstand a paper. Its draft is kept inspectable, and source disagreements and unresolved judgements remain visible to the author.

The workflow must work for people with different levels of expertise. A geneticist may want to decide every genotype and evidence statement. A non-specialist may provide a topic and several papers, then rely on the assistant to explain each decision. The tool surface cannot

assume which kind of author is present. It can perform mechanical corrections and keep a record of them, while presenting scientific interpretation and production publishing for review.

The skills help the agent follow this path, but they are instructions rather than a security boundary. An agent can still bypass them and edit a CSV directly. The same is true of the provenance log: tool-mediated overrides are recorded, while a hand edit depends on the agent calling the recording tool. These limits are important because the paper evaluates a workflow composed of tools and instructions, not a closed application that prevents every other action.

Agent hosts and MCP conventions change quickly. This paper focuses on the parts that should survive a client update. Schema answers come from the installed models. Edits can carry attribution, and source disagreements remain visible. Publication is rehearsed on a separate instance. The compiler retains authority over the artifact. Client-specific setup belongs in the repository documentation.

Limitations. The present evaluation has several limitations. It contains no creation accuracy estimate; the available fixtures are too small for a performance claim. The plugin cannot determine whether an annotation is medically correct. It can make the claim structured, attributable, and easier to review. The module authoring workflow targets GRCh38 and does not perform liftover. Local installation into Just-DNA-Lite is described as a procedure that runs in that checkout; just-module-creator does not execute the consumer on the author’s behalf. We have not compared the runtime of freely generated VCF scripts with the Just-DNA-Lite pipeline. The performance argument made here is architectural: the established path reuses normalized Parquet data and a maintained join implementation instead of generating a new analysis for each request. The companion platform paper [Anonymous, 2026] provides the runtime benchmarks.

Relationship to the companion paper. This paper describes the module authoring path. The companion platform paper [Anonymous, 2026] describes the consumer side: the Just-DNA-Lite application, its VCF processing pipeline, annotation speed benchmarks, the **just-prs** polygenic risk score engine, and the web interface. The two papers share the same module contract but describe different sides of the boundary and can be read independently.

6 Conclusion

The Just-DNA ecosystem separates two kinds of work. Module authors turn papers and databases into structured, attributable annotation knowledge. Just-DNA-Lite filters and normalizes a local genome, joins it to selected compiled modules, and produces results for inspection and reporting. A companion platform paper [Anonymous, 2026] describes that consumer side in detail.

Within the authoring side, just-module-creator lets an AI assistant draft and revise ordinary module files while the installed schema, enricher, compiler, and registry provide the shared path from draft to published artifact. The production catalog holds eight modules from five namespaces, demonstrating that the authoring and publishing workflow is functional end to end. A module can then be installed locally or consumed from the catalog without asking the language model to write another genomic pipeline. Scientific judgement belongs to the author, and sample-level computation belongs to deterministic software. The software is open-source and intended for research use only.

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A Code and data availability

- **just-module-creator**: <https://github.com/dna-seq/just-module-creator> (MCP server, Claude Code and Codex plugin manifests, skills; MIT).
- **just-dna-format**, **just-dna-compiler**, and **just-dna-enricher**: <https://github.com/dna-seq/just-dna-format> (schema, reference compiler, and enricher).
- **just-dna-registry**: production catalog at <https://module-registry.just-dna.life> and polygon at <https://module-polygon.just-dna.life>.

- **Just-DNA-Lite:** <https://github.com/dna-seq/just-dna-lite> (local VCF processing, annotation, and reporting; described in the companion platform paper [Anonymous, 2026]).
- **just-prs:** <https://github.com/dna-seq/just-prs> (polygenic risk-score computation library).

B Research use only

Annotation modules summarize published association findings. They are not clinical-grade evidence. The authoring plugin does not make individual-level predictions and does not open a genome. Just-DNA-Lite and **just-prs** process sample data for research and educational use; their reports and scores are not diagnoses or medical advice. Language that implies causation, such as “causes” or “guarantees,” is outside the scope of a module written with this plugin.